Agisoft Metashape

Processing Report

08 September 2026



Survey Data

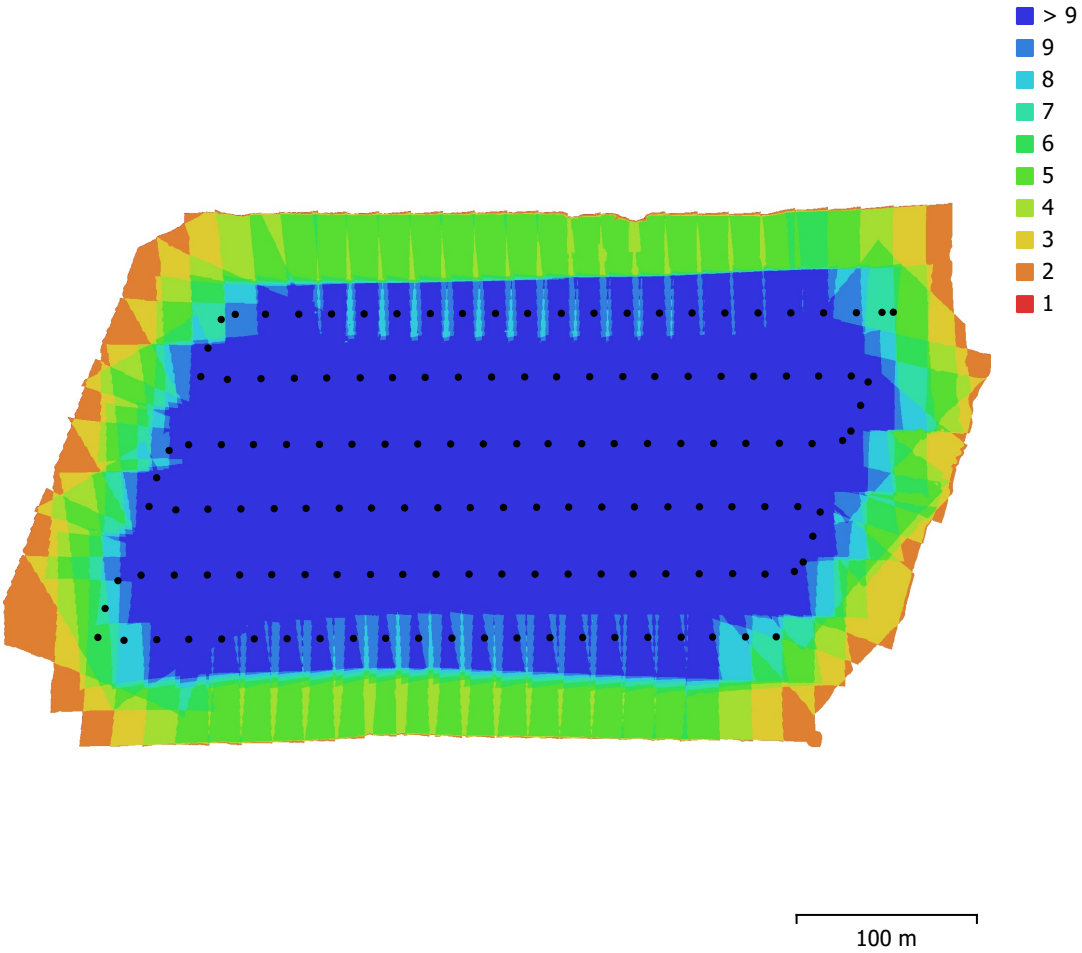


Fig. 1. Camera locations and image overlap.

Number of images:	140	Camera stations:	140
Flying altitude:	84.3 m	Tie points:	116,249
Ground resolution:	2.89 cm/pix	Projections:	379,207
Coverage area:	0.139 km²	Reprojection error:	0.61 pix

Camera Model	Resolution	Focal Length	Pixel Size	Precalibrated
ZH20T (4.5mm)	4056 x 3040	4.5 mm	1.6 x 1.6 µm	No

Table 1. Cameras.

Camera Calibration

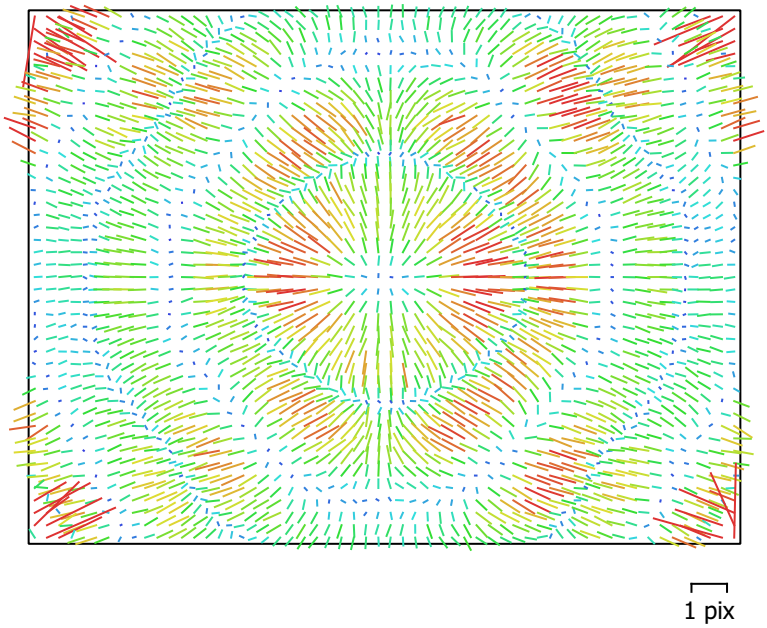


Fig. 2. Image residuals for ZH20T (4.5mm).

ZH20T (4.5mm)

140 images

Type	Resolution	Focal Length	Pixel Size
Frame	**4056 x 3040**	**4.5 mm**	**1.6 x 1.6 µm**

	Value	Error	Cx	Cy	B1	B2	K1	K2	K3	K4	P1	P2
F	**2811.66**											
Cx	**6.3911**	0.1	1.00	0.03	0.09	-0.05	-0.01	0.02	-0.02	0.02	-0.60	0.01
Cy	**13.2236**	0.084		1.00	0.05	0.08	-0.04	0.03	-0.03	0.03	0.01	-0.26
B1	**-5.08731**	0.018			1.00	-0.01	0.01	-0.02	0.02	-0.01	-0.03	-0.06
B2	**-0.190884**	0.018				1.00	-0.01	0.01	-0.00	0.01	0.03	-0.03
K1	**0.256026**	0.00022					1.00	-0.79	0.74	-0.72	-0.09	0.07
K2	**-0.810773**	0.00076						1.00	-0.99	0.96	0.01	-0.01
K3	**0.960384**	0.0015							1.00	-0.99	-0.01	0.00
K4	**-0.457444**	0.00097								1.00	0.02	-0.00
P1	**0.000210857**	5.3e-06									1.00	-0.02
P2	**-0.000768977**	5.7e-06										1.00

Table 2. Calibration coefficients and correlation matrix.

Camera Locations

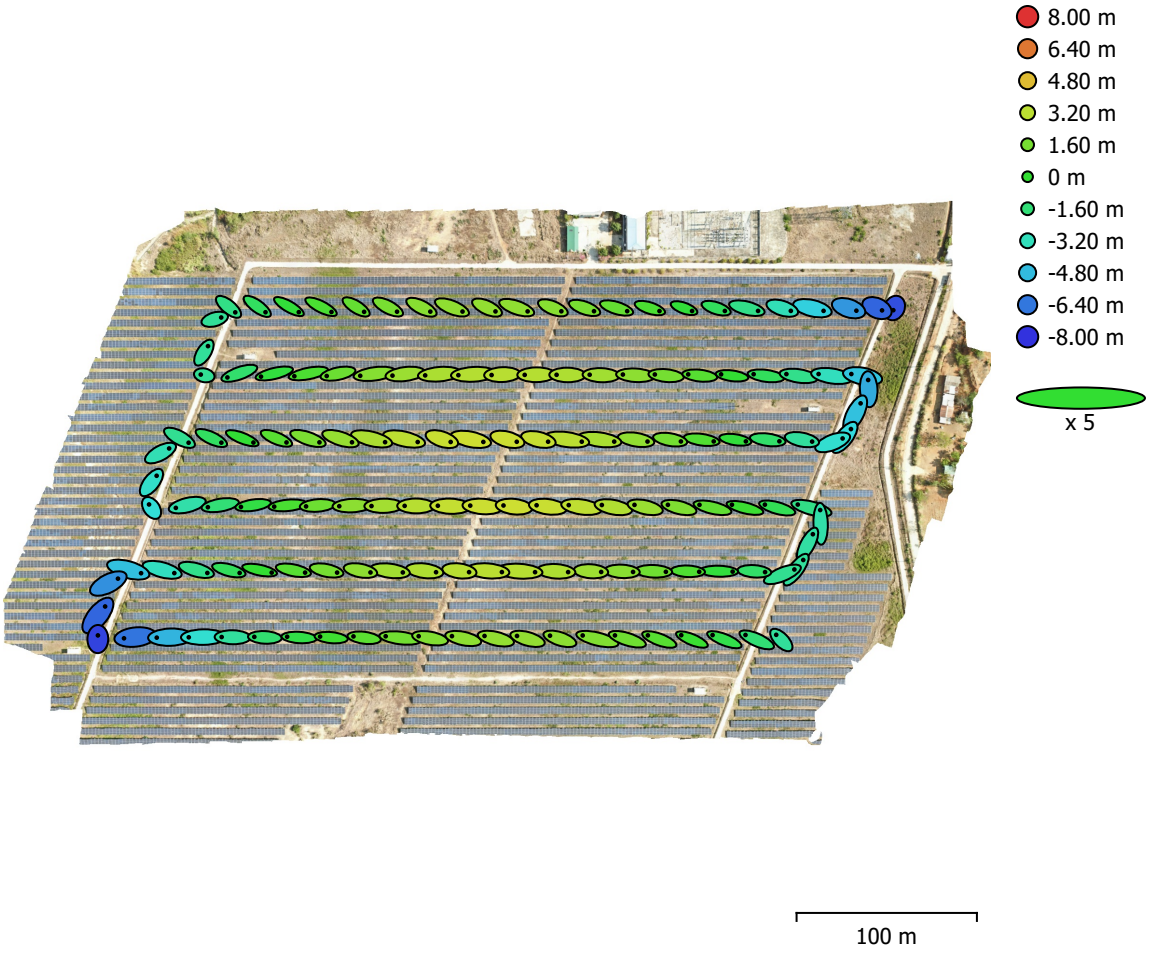


Fig. 3. Camera locations and error estimates.
Z error is represented by ellipse color. X,Y errors are represented by ellipse shape.
Estimated camera locations are marked with a black dot.

X error (m)	Y error (m)	Z error (m)	XY error (m)	Total error (m)
2.43278	0.806584	2.77692	2.56301	3.77893

Table 3. Average camera location error.
X - Longitude, Y - Latitude, Z - Altitude.

Digital Elevation Model

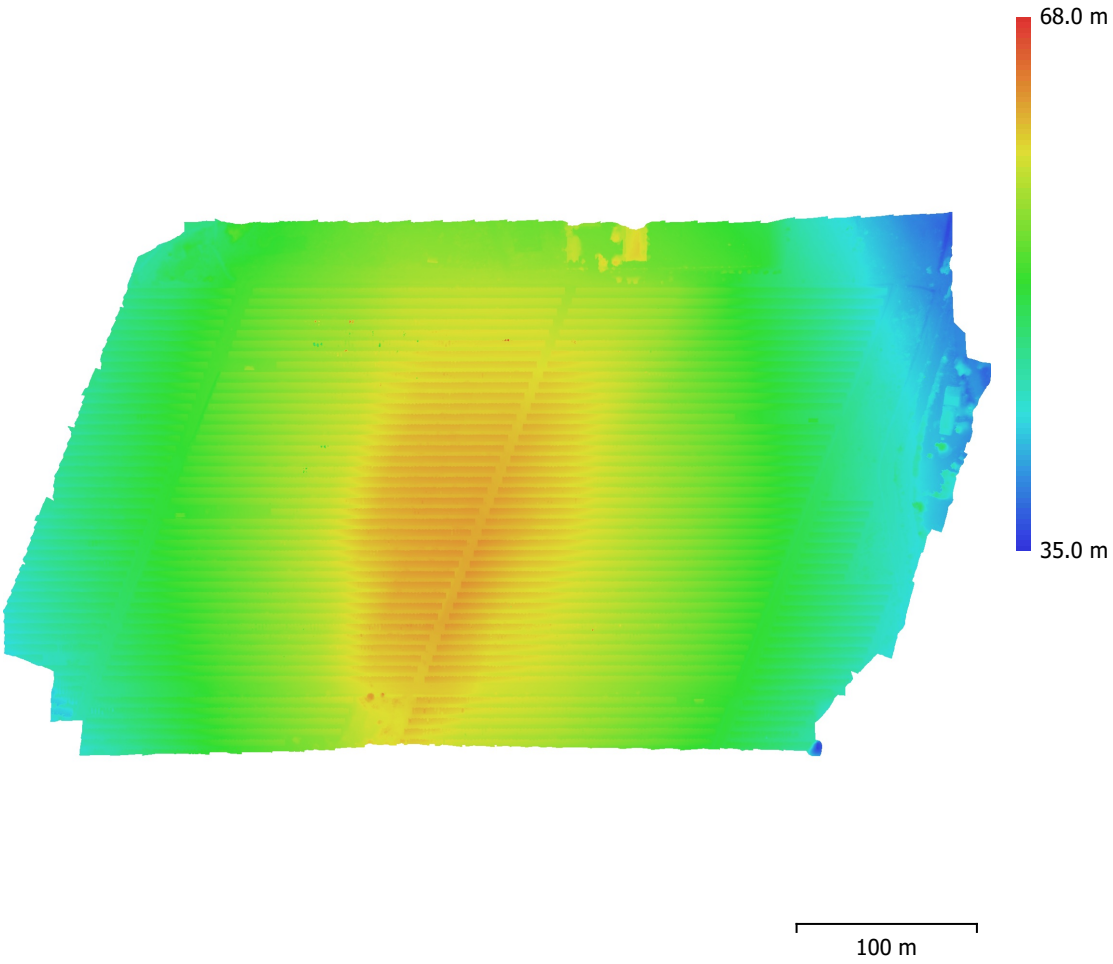


Fig. 4. Reconstructed digital elevation model.

Resolution:	6.00 cm/pix
Point density:	278 points/m²

Processing Parameters

General	
Images	140
Aligned images	140
Coordinate system	WGS 84 (EPSG::4326)
Rotation angles	Yaw, Pitch, Roll
Tie Points	
Points	116,249 of 171,046
RMS reprojection error	0.262671 (0.610475 pix)
Max reprojection error	1.40368 (4.21326 pix)
Mean key point size	2.33897 pix
Point colors	3 bands, uint8
Key points	460.36 MB
Average tie point multiplicity	3.53687
Alignment parameters	
Accuracy	High
Generic preselection	Yes
Reference preselection	Source
Key point limit	40,000
Key point limit per Mpx	1,000
Tie point limit	4,000
Exclude stationary tie points	Yes
Guided image matching	No
Adaptive camera model fitting	Yes
Matching time	3 minutes 51 seconds
Matching memory usage	396.34 MB
Alignment time	2 minutes 40 seconds
Alignment memory usage	217.54 MB
Optimization parameters	
Parameters	f, cx, cy, k1, k2, p1, p2
Adaptive camera model fitting	Yes
Exclude corners	No
Optimization time	3 seconds
Date created	2026:08:12 15:49:33
Software version	2.3.1.22580
File size	475.07 MB
Depth Maps	
Count	140
Depth maps generation parameters	
Quality	High
Filtering mode	Aggressive
Max neighbors	16
Processing time	5 minutes 54 seconds
Memory usage	2.17 GB
Date created	2026:08:12 16:09:46
Software version	2.3.1.22580
File size	630.33 MB
Point Cloud	
Points	50,564,352
Coordinate precision	1.45 cm
Point attributes	

Color	3 bands, uint8
Normal	
Point classes	
Created (never classified)	50,564,352
Depth maps generation parameters	
Quality	High
Filtering mode	Aggressive
Max neighbors	16
Processing time	5 minutes 54 seconds
Memory usage	2.17 GB
Point cloud generation parameters	
Source data	Depth maps
Processing time	9 minutes 48 seconds
Memory usage	3.59 GB
Date created	2026:08:12 16:19:37
Software version	2.3.1.22580
File size	671.48 MB
DEM	
Size	9,135 x 5,025
Resolution	6.00 cm/pix
Coordinate system	WGS 84 (EPSG::4326)
Depth maps generation parameters	
Quality	High
Filtering mode	Aggressive
Max neighbors	16
Processing time	5 minutes 54 seconds
Memory usage	2.17 GB
Reconstruction parameters	
Source data	Depth maps
Interpolation	Enabled
Processing time	20 minutes 38 seconds
Memory usage	4.92 GB
Date created	2026:08:12 16:40:29
Software version	2.3.1.22580
File size	178.36 MB
Orthomosaic	
Size	18,252 x 10,050
Resolution	3.00 cm/pix
Coordinate system	WGS 84 (EPSG::4326)
Colors	3 bands, uint8
Orthophotos	883.56 MB
Reconstruction parameters	
Blending mode	Mosaic
Surface	DEM
Enable hole filling	Yes
Enable ghosting filter	No
Processing time	3 minutes 4 seconds
Memory usage	1.10 GB
Date created	2026:08:12 16:43:05
Software version	2.3.1.22580
File size	1.12 GB
System	
Software name	Agisoft Metashape Professional
Software version	2.3.0 build 21954
OS	Windows 64 bit
RAM	15.78 GB

CPU	11th Gen Intel(R) Core(TM) i5-11400H @ 2.70GHz
GPU(s)	NVIDIA GeForce RTX 3050 Laptop GPU